Class fairness in model-heterogeneous federated learning

A measurement study, a diagnostic ladder, and a research roadmap

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Repository: github.com/Paul4805/phofl-class-fairness

Artifacts: phofl/analysis, phofl/paper, phofl_method, phofl_half

The short version

Client-FM - the usual standard deviation of per-client accuracy - hides a **44.3-point** best-to-worst class-recall spread in the PHP-FL/linear run.

FedAPEN/linear has a **52.9-point** gap, so the observation is field-wide across the two MH-FL methods tested, not universal across all FL.

A four-rung diagnostic ladder (rehearsal, confusion loss, worst-class aggregation, per-class distillation) reduced the gap by at most **7.9%** over one seed.

Step 2 is the clearest mechanism: dog-to-cat errors fell 23%, but cat recall fell 16%, and the combined dog+cat error rose 0.657 -> 0.683.

class_FM is a **reporting metric**, not a training objective. No method contribution is claimed.

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Executive summary

This report documents a measurement project that started from the PHP-FL paper and ended with a two-axis fairness critique plus a negative-result diagnostic ladder. The central claim is simple: a single client-level dispersion number is not a complete fairness report for model-heterogeneous federated learning (MH-FL). A model can look well balanced across clients while leaving a large semantic class spread.

On the half-scale CIFAR-10 setup, PHP-FL/linear has client-FM 0.0785 and a class gap of 44.3 points: frog recall is 69.4% while dog recall is 25.1%. The single best client-FM run (PHP-FL/entropy, FM=0.0763) is slightly worse on the class axis at 45.6 points. FedAPEN/linear reaches 52.9 points. Because FedAPEN crosses the pre-registered 30-point scope threshold, the finding is field-wide across the two MH-FL methods in this study, but it is not a universal claim about all FL.

The five runs also order differently on the two axes: FedAPEN is 3rd by client-FM but 5th by class gap, while Standalone/linear is 4th by client-FM but 1st by class gap. This is an observation on five runs, not a rank-correlation claim.

The diagnostic ladder was designed to test whether the gap is reducible. Four standard interventions were run: class-balanced rehearsal, confusion-targeted loss, worst-class-weighted aggregation, and per-class distillation. The largest reduction was 7.9%, from rehearsal. Step 2 is the most informative: it moved dog-to-cat errors in the intended direction but shifted the error into cat, leaving the combined dog+cat error slightly higher. This is consistent with an information-theoretic floor at CIFAR-10 resolution and argues for class_FM as a reporting metric, not an optimization target.

1. The PHP-FL paper in brief

PHP-FL addresses fairness in model-heterogeneous federated learning when clients have different participation probabilities. The setting is deliberately heterogeneous: four CNN architectures are assigned round-robin to 20 clients, so there is no single global model that can simply be averaged across all clients. The paper's fairness target is client-level: reduce the dispersion of per-client test accuracy under participation inconsistency.

1.1 Method components

- **Private and auxiliary models.** Each client has a private model w_i and a local copy g_i of a shared auxiliary model. The server maintains the global auxiliary model G. The private model carries personalization; the auxiliary model carries shared knowledge.
- **DEAL: dual-end alignment.** The local objective combines cross-entropy with two alignment terms: an MMD term between the private and shared backbone features, and a KL term between their logits. An ensemble loss mixes the two predictions. The intent is to align representations and outputs at both ends while preserving personalization.
- **ISPU: individual-specific parameter update.** Each client computes a participation ratio alpha from its cumulative participation count N_i(t). The top-alpha parameters of its auxiliary model, ranked by L1 norm, are overwritten with the corresponding global parameters. In the repository the schedule is `alpha = tau * sigmoid(k, (N_i(t)+1)/(t+1))` with `sigmoid(k,x)=1/(1+exp(k(x-0.5)))`. Rare clients therefore receive a larger alpha and more global knowledge; frequent clients receive less.
- **Server aggregation.** Active clients upload their auxiliary models; the server takes a plain mean over active clients; active clients then apply their ISPU masks. The private models are not aggregated.
- **Evaluation.** AM is the mean per-client accuracy; FM is the standard deviation of per-client accuracy. Participation patterns are uniform, normal(mu, sigma), and a shuffled linear/arithmetic schedule.

1.2 What the paper claims

The paper claims that the combination of dual-end alignment and individual-specific updates improves client fairness under participation-probability inconsistency in heterogeneous environments. The primary evidence is the FM/AM table on CIFAR-10 and related datasets. The method is evaluated as a client-fairness method; class-level recall is not part of the reported metric.

1.3 What this project tests

This project does not re-derive or replace PHP-FL. It asks a complementary measurement question: if a method improves client-FM, does it also improve the class axis? The repository's PHP-FL code is used as the reference implementation; the class-fairness analysis is layered on top of the same training and evaluation logs.

2. Project arc: what was planned, what was run

The project began with a much larger plan: reproduce the paper's CIFAR-10 normal-pattern baseline (AM 66.94, FM 8.07), then run aggregation, participation, cold-start, convergence, efficiency, membership-inference, and baseline-porting experiments. The scope was reduced before the full baseline matrix ran. As a result, this repository does **not** contain a successful reproduction of Table 2, and no claim of exact paper reproduction is made.

The work that was actually completed followed this arc:

- **Phase A/B/C (Fashion-MNIST smoke).** Fairness-metric recomputation, aggregation reweighting, and correlated-participation comparisons on 10 clients, 40 rounds, two backbones. These are diagnostic smoke runs, not CIFAR-10 claims.
- **Participation branch closed.** The C escalation was pre-registered, corrected with Holm, and repeated on the pinned stack. The linear-vs-entropy pattern effect did not survive correction, and the half-scale run failed the pre-registered thresholds. The participation hypothesis is closed and is not revived.
- **Class-fairness pivot.** The analysis moved to the two-axis object: client x class. Five CIFAR-10 half-scale runs were analyzed: PHP-FL linear/entropy, FedAPEN linear, and Standalone linear/entropy.
- **FedAPEN scope test.** FedAPEN/linear was added as the second MH-FL method. Its 52.9-point class gap promoted the finding to field-wide across the two methods tested.
- **Diagnostic ladder.** Four interventions were tested one at a time to see whether the class gap is reducible. None closed it; the maximum reduction was 7.9%.

The most important consequence of the arc is methodological: the project changed from trying to build a better method to measuring whether the reported fairness metric captures the failure it claims to address. That change is the paper.

3. Experimental setup and metric definitions

3.1 Half-scale CIFAR-10 setup

Item	Value
Dataset	CIFAR-10, 10 clients, Dirichlet(alpha=0.5)
Per-client data	2,500 train / 500 test
Training	60 rounds, 1 local epoch, batch 64 (32 for heavy backbones)
Models	Four CNNs: GoogLeNet, DenseNet-121, EfficientNet-B1, ResNet-18
Auxiliary	GoogLeNet, plain mean aggregation
Seed	0 (one seed)
Stack	torch 2.2.2+cu121, numpy 1.26.4, pinned
Evaluation	Full 1,000-sample CIFAR-10 test set per class

This is a compute-bounded half-scale setting: roughly 8x less training than the paper's full CIFAR-10 setting. Absolute AM/FM are not comparable to the paper. The linear-vs-entropy contrast and the class-axis geometry are the usable outputs.

3.2 Metrics

- **AM:** mean per-client accuracy. **FM:** population standard deviation of per-client accuracy. In the artifacts, `AM_std` is numerically FM, not a seed-level standard error.
- **Class gap:** best per-class recall minus worst per-class recall on the full test set. **class_std:** population standard deviation of the 10 per-class recalls.
- **Worst-class recall:** minimum per-class recall. This is the tail metric that a client-level average can hide.
- **Support filter:** per-(client, class) cells with fewer than 10 test samples are excluded from worst-cell and support-filtered calculations. Each run has 34/100 such cells (170 across five runs; 45 have zero samples). The global class headline uses the full 1,000 samples per class and is not affected.
- **class_FM:** standard deviation over classes of per-class mean recall. It is the class-axis analogue of client-FM and is proposed as a reporting metric, not a training objective.
- **AM-normalized gap:** class gap divided by AM. This asks whether a smaller raw gap is simply a lower-accuracy effect. The ceiling-normalized variant divides by 1 - worst-class recall.

4. First measurement phase: A/B/C

The first phase was a Fashion-MNIST smoke study: 10 clients, 40 rounds, 1 seed, GoogLeNet and ResNet-18, 1,000 train / 200 test per client. It was designed to test the measurement pipeline and the participation branch, not to make CIFAR-10 claims.

4.1 A: fairness is more than std

Metric	Plain PHP-FL baseline
AM	0.5175
FM	0.1607
min accuracy	0.2400
10th percentile	0.2985
CVaR10	0.2400
CV	0.3105
Jain	0.9120
Spearman rho(p, accuracy)	-0.0061

The original A narrative expected PHP-FL to have the lowest FM but not the best tail. In this smoke baseline the FM winner also won min/CVaR, and rho was not positive. The narrative was not confirmed on this run. This is a negative result about the baseline, not about the CIFAR-10 paper.

4.2 B: aggregation reweighting was a null result

Aggregation	AM	FM	min
plain	0.5175	0.1607	0.240
inv_p	0.5160	0.1628	0.260
inv_N	0.5105	0.1615	0.260

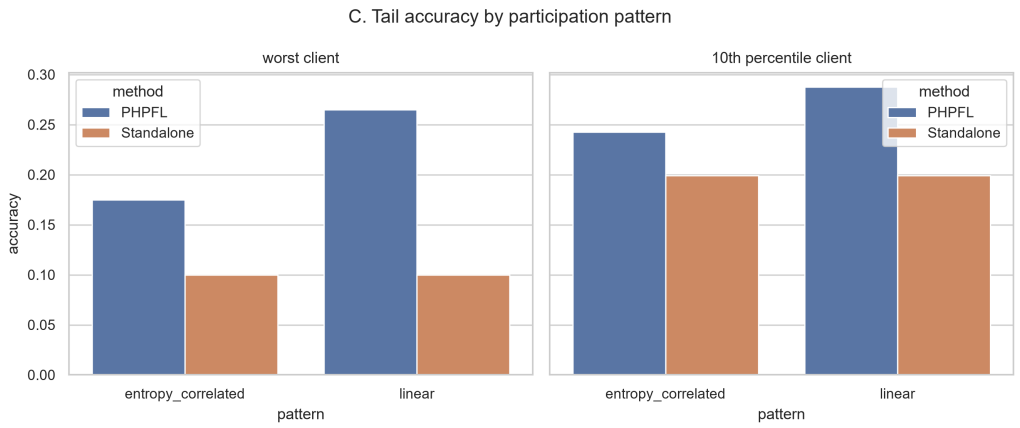
The tail minimum improved by about 2 points under inv_p and inv_N, but the paired Wilcoxon tests on client accuracy were null: plain vs inv_p p=0.922, plain vs inv_N p=0.469, inv_p vs inv_N p=0.570. The aggregation branch was written up as a null result and moved past.

4.3 C: correlated participation and the closed branch

Pattern	Method	AM	FM	min	rho(p,acc)
linear	PHPFL	0.5040	0.1439	0.265	-0.758
entropy	PHPFL	0.5120	0.1805	0.175	+0.721
linear	Standalone	0.4605	0.1856	0.100	-0.794

The entropy-correlated sampler raised mean accuracy slightly but increased dispersion and collapsed the tail. The raw linear-vs-entropy dispersion test was p=0.041; after Holm correction it became p=0.164 and lost significance. PHP-FL vs Standalone under entropy survived Holm (p=0.0098), but that is a method contrast, not a pattern effect. The pinned-stack smoke run preserved the sign but not the significance of the dispersion effect.

The C escalation then ran at CIFAR-10 10 clients / 60 rounds. It failed the pre-registered thresholds: FM increase -2.84% (GO required >25%), min accuracy +4.09 points (GO required a >5-point drop), rho difference 0.170 (GO required >1.0). C is closed. This report does not revive the participation hypothesis.



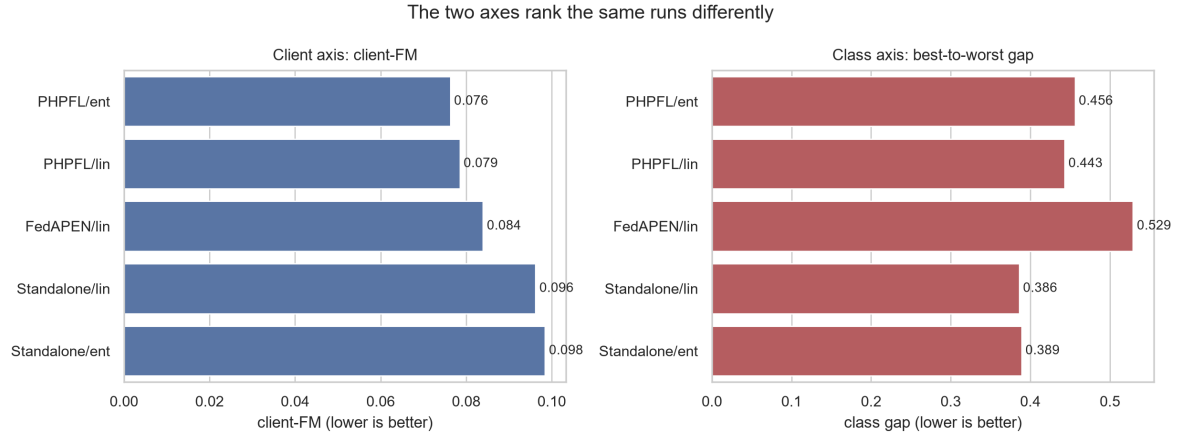
Fashion-MNIST smoke: entropy-correlated participation collapses the tail. This is a smoke diagnostic, not a CIFAR-10 result.

5. Two-axis fairness on CIFAR-10

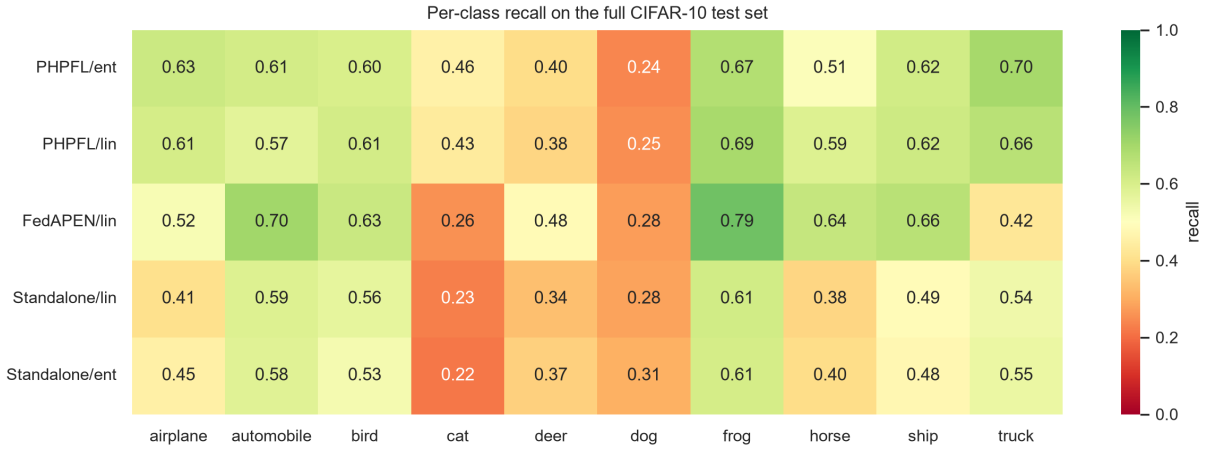
The central measurement result is the client-by-class matrix. Client-FM is the row-mean dispersion; class_std and worst-class recall are the column summaries. The five-run table below is the core evidence.

Run	AM	client-FM	class_std	class gap	worst class	worst recall
PHPFL/ent	0.6308	0.0763	0.1349	0.4560	dog	0.2400
PHPFL/lin	0.6320	0.0785	0.1334	0.4430	dog	0.2510
FedAPEN/lin	0.6209	0.0839	0.1698	0.5290	cat	0.2560
Standalone/lin	0.5429	0.0962	0.1281	0.3860	cat	0.2290
Standalone/ent	0.5467	0.0985	0.1196	0.3890	cat	0.2180

The conservative headline is 44.3 points in PHP-FL/linear: frog 69.4%, dog 25.1%. The single best client-FM run is PHP-FL/entropy (0.0763) at 45.6 points. FedAPEN/linear is 52.9 points. The failure is not a single-class artifact: dog, cat, deer, and horse form a broad weak region.



Client-FM and class gap on the same five runs. The bars are ordered by client-FM on the left; the class axis on the right does not follow that order.



Per-class recall. Rows are runs; columns are classes. The weak region is dog/cat/deer/horse, not one isolated cell.

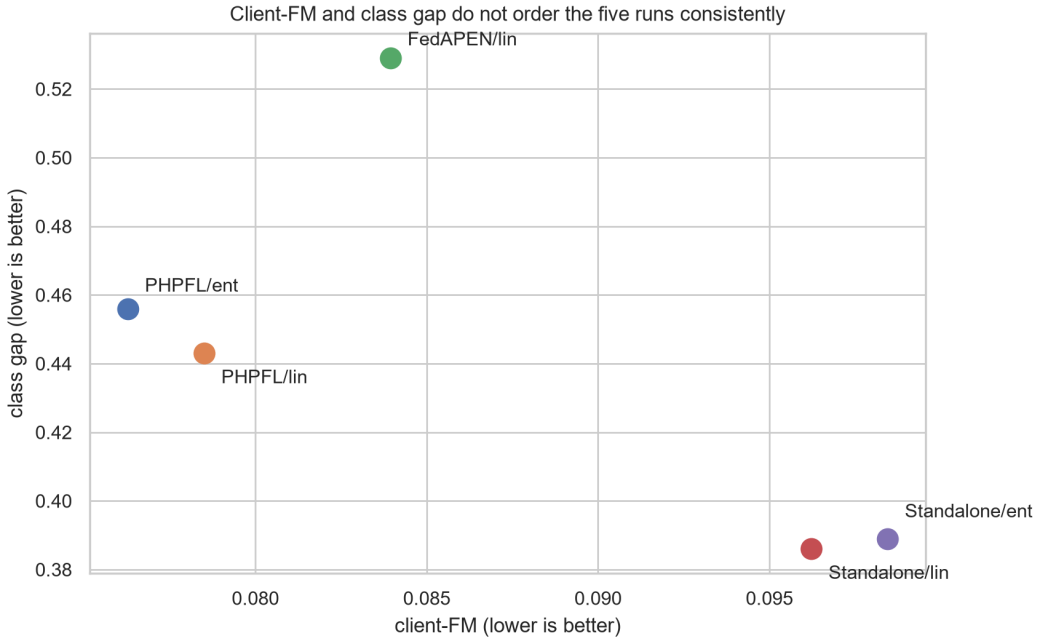
5.1 Support filtering

Per-client class cells are sparse: 34/100 cells per run have fewer than 10 test samples, 170 cells across five runs, including 45 zero-sample cells. These are listed in `low_support_cells.csv` and excluded from the worst-cell calculation. The worst supported cell is still 0.0 (client 0, cat, 11 samples) in every run. It is sparse but above threshold and is not the headline. Computing class means only from supported cells gives a 51.3-point gap for PHP-FL/linear, a 45.0-point gap for PHP-FL/entropy, and a 41.6-point gap for FedAPEN; 44.3 is conservative.

5.2 Rankings disagree

Run	client-FM	FM rank	class gap	gap rank
PHPFL/ent	0.0763	1	0.456	4
PHPFL/lin	0.0785	2	0.443	3
FedAPEN/lin	0.0839	3	0.529	5
Standalone/lin	0.0962	4	0.386	1
Standalone/ent	0.0985	5	0.389	2

The two metrics order the five runs differently. FedAPEN is 3rd by client-FM but 5th by class gap; Standalone/linear is 4th by client-FM but 1st by class gap. This is an observation on five runs, not a rank-correlation claim: n=5 is too small to test whether the disagreement is systematic.



The same five runs in the client-FM / class-gap plane. The ordering is not monotone; the two axes disagree.

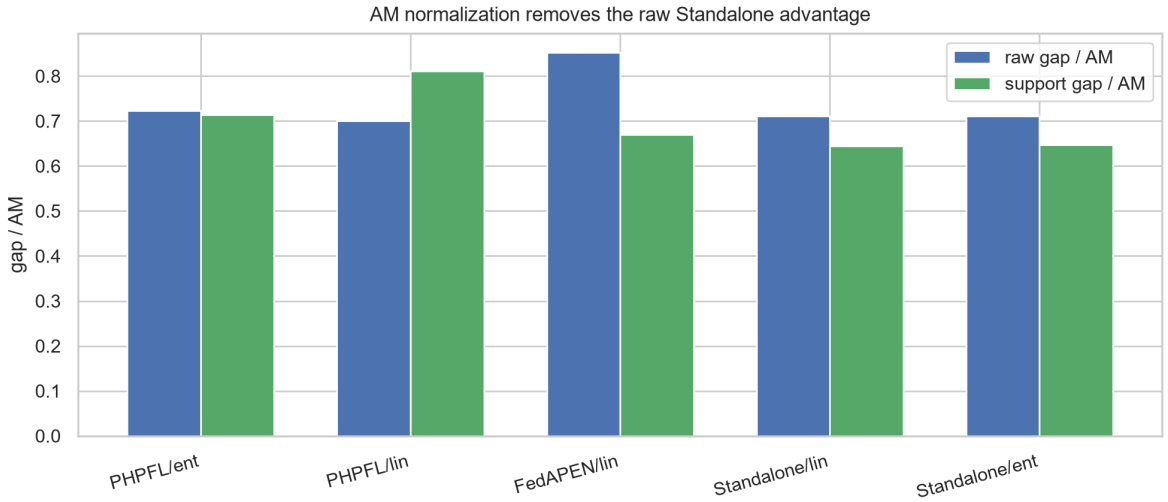
5.3 Pattern invariance

Class-std is roughly invariant to the participation pattern within a method: PHP-FL 0.1334 -> 0.1349 (change 0.0015) and Standalone 0.1281 -> 0.1196 (change 0.0084). The class gap changes by about 1.3 points for PHP-FL and 0.3 points for Standalone. The class imbalance is structural; the linear-vs-entropy sampler does not materially create or remove it.

5.4 AM normalization and the Standalone caveat

Run	AM	raw gap/AM	support gap/AM
PHPFL/ent	0.6308	0.723	0.713
PHPFL/lin	0.6320	0.701	0.811
FedAPEN/lin	0.6209	0.852	0.670
Standalone/lin	0.5429	0.711	0.645
Standalone/ent	0.5467	0.712	0.648

Standalone has a smaller raw class gap (38.6 vs 44.3), but its AM is much lower. After normalizing the raw gap by AM, Standalone's advantage disappears (0.711 vs 0.701 for PHP-FL/linear). We therefore do not claim a Standalone class-fairness advantage. The support-filtered gap/AM is lower for Standalone, but it is reported only as a sensitivity, not the headline. The PHP-FL vs FedAPEN inversion is AM-independent: FedAPEN is 3rd by client-FM and 5th by gap/AM.



AM-normalized gap. The raw Standalone advantage is removed by AM normalization; the PHP-FL/FedAPEN inversion survives.

5.5 FedAPEN and the field-wide scope

FedAPEN/linear has a 52.9-point raw class gap (cat 25.6%, frog 78.5%) and a 41.6-point support-filtered gap. The pre-registered rule was: >30 points promotes to field-wide, <20 narrows to PHP-FL, 20-30 is mixed. At 52.9, the claim is field-wide across the two MH-FL methods tested (PHP-FL and FedAPEN), not universal across all FL.

6. The diagnostic ladder

If a standard intervention could remove the class gap, the metric critique would be less interesting. The ladder tested four one-at-a-time interventions under the same half-scale configuration. The baseline is the existing PHP-FL/linear run.

Rung	Intervention	AM	FM	gap	reduction	dog	frog
0	PHP-FL/linear baseline	0.6320	0.0785	0.443	-	0.251	0.694
1	class-balanced rehearsal (50/class)	0.6099	0.0854	0.4080	+7.9%	0.3200	0.7280
2	confusion-targeted loss (top-3, lambda=0.1)	0.6346	0.0900	0.4500	-1.6%	0.2700	0.7200
3	worst-class-weighted aggregation (gamma=5)	0.6312	0.1002	0.4310	+2.7%	0.2920	0.7230
4	per-class distillation (T=1.0)	0.6322	0.0944	0.4240	+4.3%	0.2820	0.7060

No rung reached the 30% closure threshold. The maximum reduction was 7.9% from rehearsal. The ladder is therefore a negative result about reducibility, not a failed method paper.



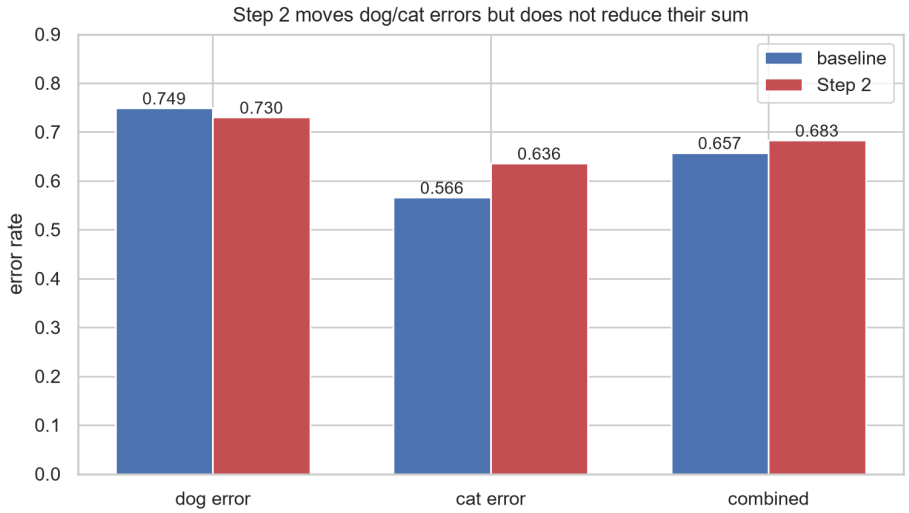
Ladder gaps and relative reduction. The 30% closure threshold is dashed; no rung approaches it.

6.1 Rehearsal: broad improvement, preserved gap

Step 1 stored 50 samples per class and mixed them 1:1 with fresh batches. Dog recall rose from 0.251 to 0.320 (+6.9 points) and frog from 0.694 to 0.728 (+3.4 points). The gap fell only from 0.443 to 0.408 (7.9%). Both ends improved, but the improvement was proportional rather than redistributive; the normalized gap barely moved.

6.2 Confusion loss: the zero-sum result

Step 2 added a margin penalty on each client's top-3 validation confusion pairs (lambda=0.1, margin 1.0). Dog-to-cat errors fell from 0.249 to 0.191 (-23%). But cat recall fell from 0.434 to 0.364 (-16%), and the combined dog+cat error rate rose from 0.657 to 0.683. The loss moved the decision boundary rather than removing the joint error. This is the clearest evidence for an information-theoretic floor at 32x32 CIFAR-10 resolution.



Step 2 shifts dog/cat errors: dog improves, cat worsens, the combined error does not fall.

6.3 Worst-class aggregation

Step 3 used weights exp(-gamma * recall_i(c_worst)) with gamma=5. The globally worst class was airplane early and dog later. The formula gives the largest weight to clients with low recall on the worst class, i.e. the clients least able to represent it. The gap fell only to 0.431 (2.7%); dog recall reached 0.292. Aggregation reweighting alone does not fix a representation problem.

6.4 Per-class distillation

Step 4 weighted DEAL's KL term by exp(r_i(c)/T) with T=1.0. Dog recall rose to 0.282, frog to 0.706, and the gap fell to 0.424 (4.3%). The best dog architecture was EfficientNet-B1 before and after; its mean dog recall rose from 0.599 to 0.629, so the best-dog architecture did not change. This is the strongest mechanism-specific result, but still far from closure.

7. What is interesting

- **1. A 44.3-point class gap in a good client-FM run.** Client-FM 0.0785 coexists with frog 69.4% and dog 25.1%. The best client-FM run is worse on the class axis at 45.6 points. A leaderboard sorted by FM would hide the failure.
- **2. FedAPEN makes the finding field-wide across two MH-FL methods.** Its gap is 52.9 points, above the pre-registered 30-point scope threshold. This is not a universal FL claim; it is field-wide across PHP-FL and FedAPEN.
- **3. The two axes order the runs differently.** FedAPEN is 3rd by client-FM but 5th by class gap; Standalone/linear is 4th by client-FM but 1st by class gap. With five runs this is an observation, not a systematic correlation.
- **4. The class bias is structural.** Changing linear to entropy changes class-std by 0.0015 for PHP-FL and 0.0084 for Standalone. Participation is not the main driver of the class axis in these models.
- **5. Four standard interventions do not close the gap.** Rehearsal, confusion loss, worst-class aggregation, and per-class distillation reduce it by at most 7.9% over one seed. The gap is not a simple optimization failure.
- **6. The confusion loss is zero-sum.** Dog-to-cat errors fall 23%, cat recall falls 16%, and the combined error rises. That is an error-allocation signature, consistent with an information-theoretic floor at CIFAR-10 resolution.
- **7. Standalone's raw gap advantage is an AM effect.** Its gap is smaller (38.6 vs 44.3), but gap/AM is 0.711 vs 0.701 after normalization. The raw claim is dropped; the support-filtered variant is a sensitivity, not a headline.

The most interesting single result is not the 44-point number by itself. It is the combination of the rank disagreement and the zero-sum confusion result: the class axis is a real, separate fairness object, and one of the most natural ways to attack it moves errors rather than removing them.

8. Limitations and what is not established

- **No Table 2 reproduction.** The original 66.94/8.07 baseline assertion was not executed after the scope change. This repository does not claim to reproduce the paper's full CIFAR-10 numbers.
- **Half-scale only.** 10 clients, 60 rounds, one seed, four CNNs, one local epoch, 2,500/500 per client. Absolute AM/FM are not comparable to the paper. Only the linear-vs-entropy contrast and the class-axis geometry are usable.
- **One seed.** No seed-level uncertainty is claimed for class-std, class gap, worst-class recall, or the rank order.
- **Five runs are not a field survey.** The field-wide scope covers two MH-FL methods (PHP-FL and FedAPEN). FedFV, FedHEAL, FedAWE, and FedAU were not ported; the original E8 branch was dropped.
- **Support sparsity.** Per-client class cells are sparse. The 10-sample threshold is a pre-declared compromise; the global class headline uses the full test set, but the client-cell tail is still noisy.

- **Descriptive, not causal.** The analysis does not identify why dog/cat/deer/horse are weak. The information-floor interpretation is a hypothesis motivated by Step 2, not a proven mechanism.

9. Future directions

The project is now a measurement paper. The most valuable next step is not another method; it is to test the mechanism behind the zero-sum result and to make the reporting practice standard.

9.1 Primary: resolution sweep and pairwise error budget

The Step 2 result suggests that dog and cat are confusable at 32x32. A clean test is to measure the full 10x10 confusion matrix at 32, 64, and 96 pixels on a small centralized probe. If the joint dog+cat error stays flat at 32 but falls at higher resolution, the floor is resolution-limited. If it stays flat even at higher resolution, the ambiguity is in the labels/data rather than the input resolution.

The same analysis should be repeated for other class pairs (deer/horse, automobile/truck). If targeted losses always move error between directions without reducing the pair sum, the claim becomes stronger: some class gaps are an error-allocation problem, not an accuracy problem.

9.2 Zero-training: architecture decomposition

The committed per-client logs already contain architecture labels and per-class recalls. A zero-training analysis can test whether the class gap is concentrated in specific architectures. EfficientNet-B1 was the best dog architecture (mean dog recall 0.599 baseline, 0.629 in Step 4); if other architectures contribute most of the cat/dog errors, the gap is partly an inductive-bias problem rather than federation.

9.3 Multi-seed and full-scale confirmation

A second seed and a full-scale CIFAR-10 audit are needed before any universal claim. The paper-scale audit would use 20 clients, 3 local epochs, and 3 seeds. This is a future run; it was not executed. The half-scale one-seed results should be treated as a diagnostic, not a final measurement.

9.4 More MH-FL methods

The field-wide scope currently rests on two methods. Porting FedFV, FedHEAL, FedAWE, and FedAU into the same heterogeneous setting would test whether class-FM is a general problem or specific to PHP-FL/FedAPEN. This is measurement, not a new method.

9.5 Reporting standard

The practical recommendation is a three-number report plus the matrix: client-FM, class-FM, worst-class recall, and the client-by-class matrix or a link to it. class_FM should be reported, not optimized. A model that wins client-FM but loses class-FM should not be presented as unambiguously fairer.

10. Reproducibility and artifact map

The public repository contains the paper source, analysis code, result CSVs, structured per-run logs, and figures. Large artifacts are excluded.

Path	Contents
phofl/paper/	Abstract through conclusion (sections 00-07)
phofl/analysis/	Client x class matrices, support filtering, AM normalization, ranks
phofl_method/	Ladder per-step CSVs, findings, LADDER_SUMMARY.md
phofl_half/results/	Half-scale result CSVs
phofl_half/logs/	Per-run round metrics and histories
phofl/results/, phofl/logs/	Earlier A/B/C scoped results
phofl/figures/	300 DPI figures
phofl_experiments.ipynb	Resumable A/B/C notebook
phofl_lab.py	Runtime harness
analyze_two_axis.py	Rebuilds the 2-D analysis
run_step1.py ... run_step4.py	Diagnostic ladder

Excluded: third_party clones, generated CIFAR-10 partitions, checkpoints, run_store, phofl_pinned, and .venv-pinned. DATA.md documents regeneration. The repository is at github.com/Paul4805/phofl-class-fairness; release v0.1.0 is tagged.

10.1 Key commands

Rebuild the committed analysis without training:

```
python analyze_two_axis.py
python make_final_table.py
python -m pytest -q
```

One-line conclusion

Client-FM is a useful but incomplete fairness report. The class axis is a real, structural second dimension; it survives a participation change, it ranks methods differently, and four standard interventions do not close it. Report class_FM and worst-class recall alongside client-FM, but do not treat class_FM as an objective.

References

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